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Gawai et al.

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(54) **METHODS FOR FORMING CONDUCTIVE VIAS, AND ASSOCIATED DEVICES AND SYSTEMS**

(71) Applicant: **Micron Technology, Inc.**, Boise, ID (US)

(72) Inventors: **Trupti D. Gawai**, Boise, ID (US); **David S. Pratt**, Meridian, ID (US); **Ahmed M. Elsied**, Boise, ID (US); **David A Kewley**, Boise, ID (US); **Dale W. Collins**, Boise, ID (US); **Raju Ahmed**, Boise, ID (US); **Chelsea M. Jordan**, Boise, ID (US); **Radhakrishna Kotti**, Meridian, ID (US)

(73) Assignee: **Micron Technology, Inc.**, Boise, ID (US)

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H01L 21/768 (2006.01)
H01L 23/522 (2006.01)
H01L 23/528 (2006.01)

(52) **U.S. Cl.**
CPC .. **H01L 21/76883** (2013.01); **H01L 21/76816** (2013.01); **H01L 23/5226** (2013.01); **H01L 23/5283** (2013.01)

(58) **Field of Classification Search**
CPC H01L 21/76883; H01L 21/76816; H01L 23/5226; H01L 23/5283
See application file for complete search history.

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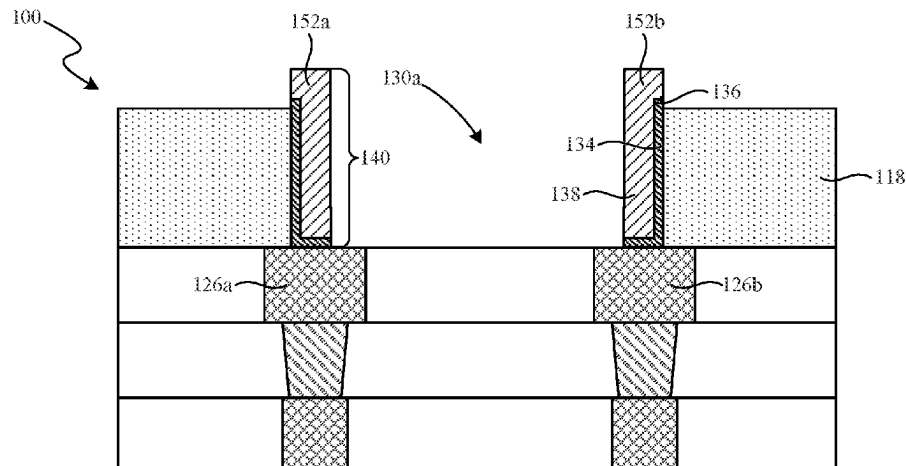
Primary Examiner — Karen Kusumakar

(74) *Attorney, Agent, or Firm* — Dorsey & Whitney LLP

(57) **ABSTRACT**

Methods of manufacturing semiconductor devices, and associated systems and devices, are disclosed herein. In some embodiments, a method of manufacturing a semiconductor device includes forming an opening in an electrically insulative material at least partially over a first electrically conductive feature and a second electrically conductive feature. The method can further include forming a ring of electrically conductive material around a sidewall of the insulative material defining the opening, wherein the ring of electrically conductive material includes (a) a first via portion over the first electrically conductive feature, (b) a second via portion over the second electrically conductive feature, and (c) connecting portions extending between the first and second via portions. Finally, the method can include removing the connecting portions of the ring of electrically

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conductive material to electrically isolate the first via portion from the second via portion.

20 Claims, 10 Drawing Sheets

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continuation of application No. 17/136,287, filed on Dec. 29, 2020, now Pat. No. 11,515,204.

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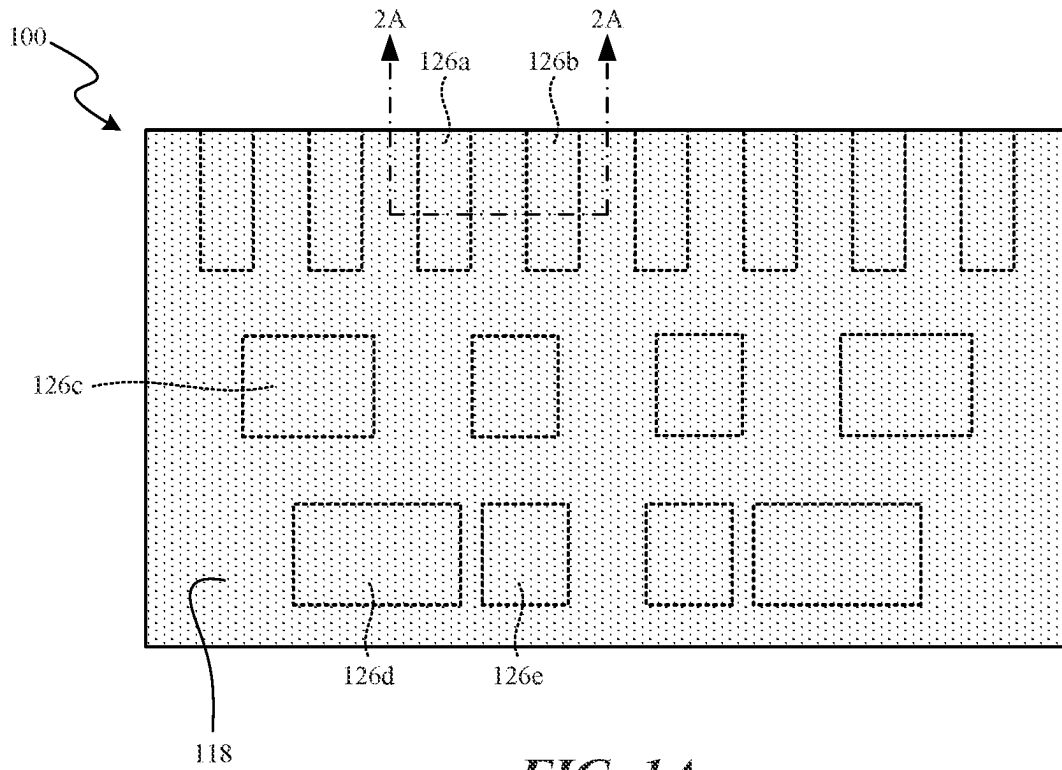


FIG. 1A

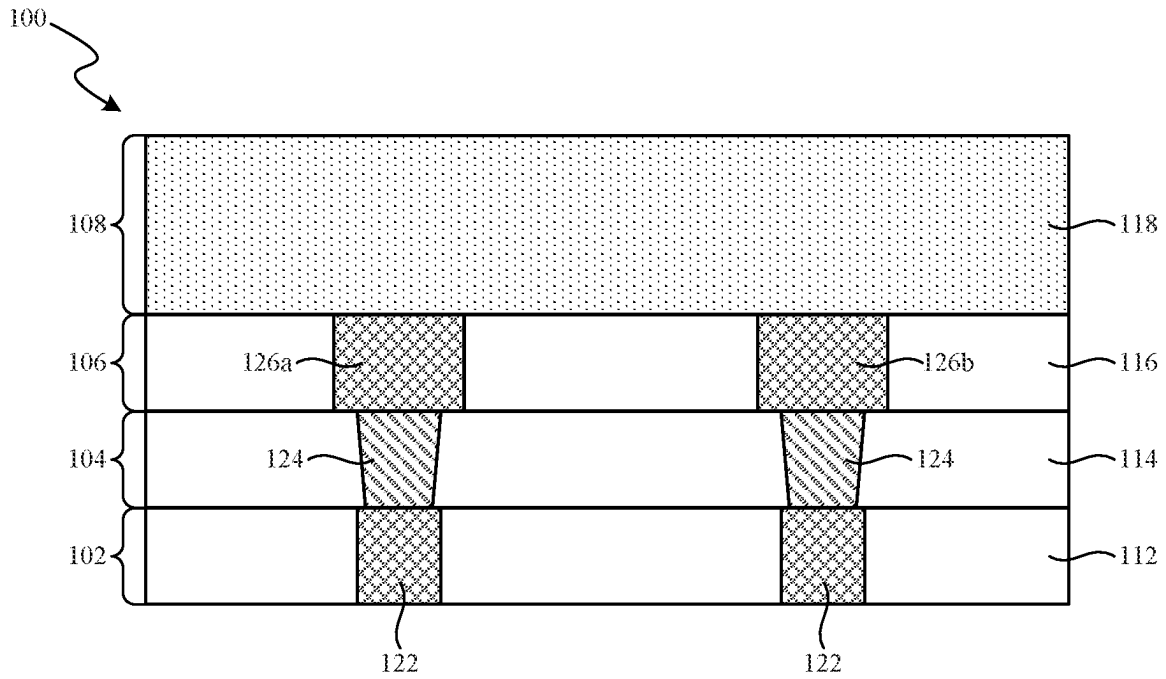


FIG. 2A

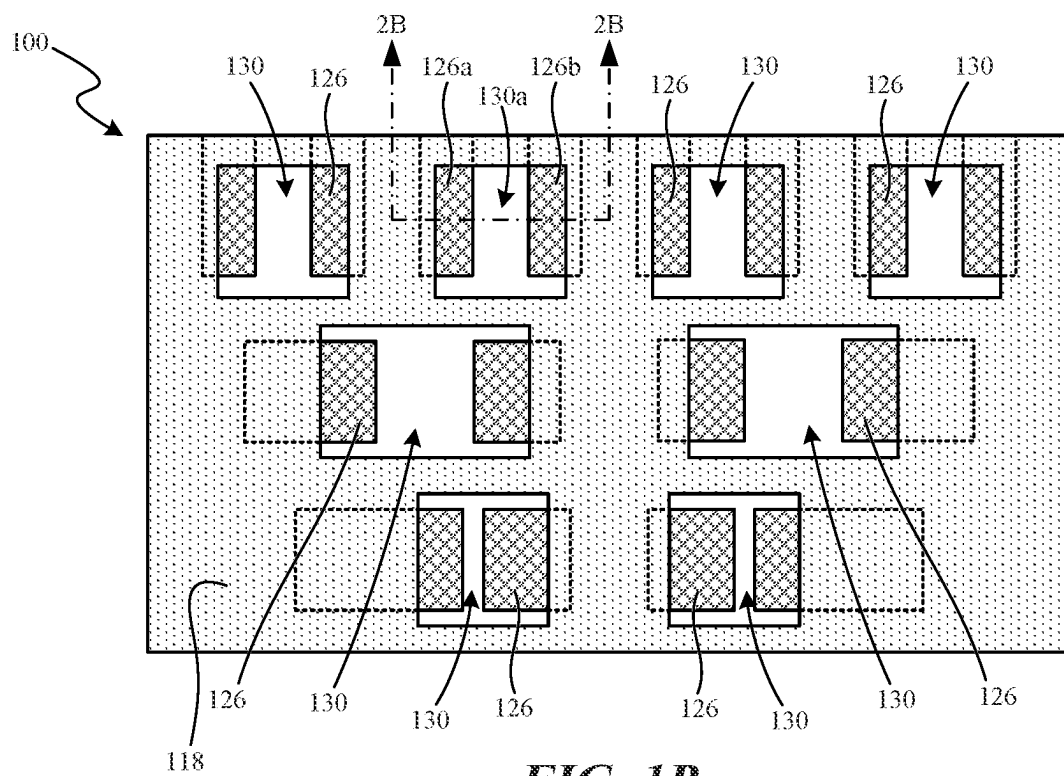


FIG. 1B

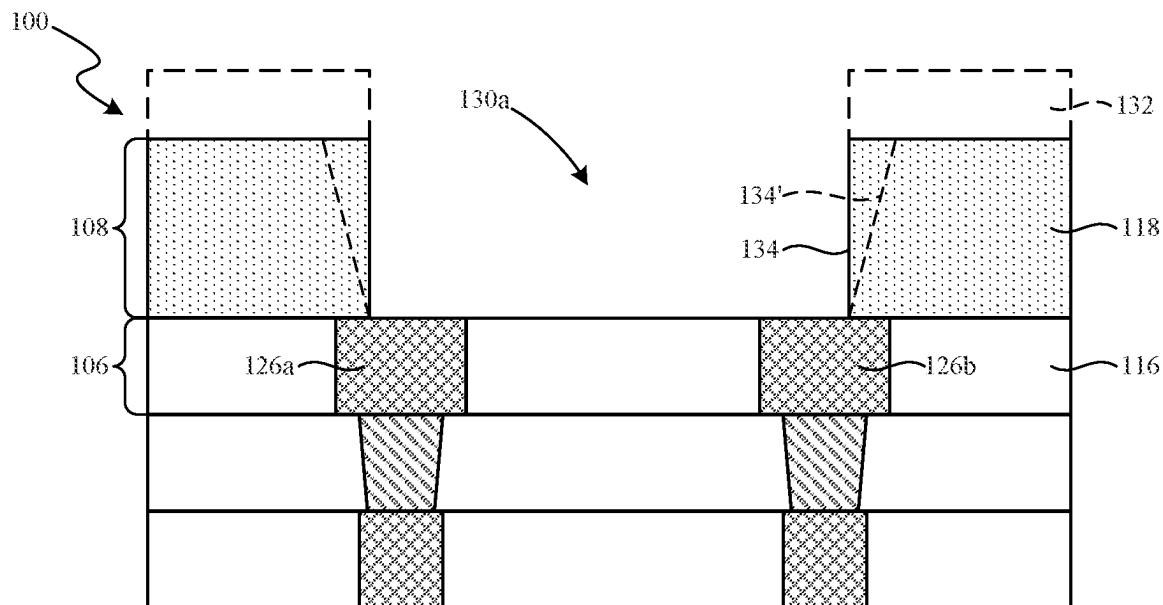


FIG. 2B

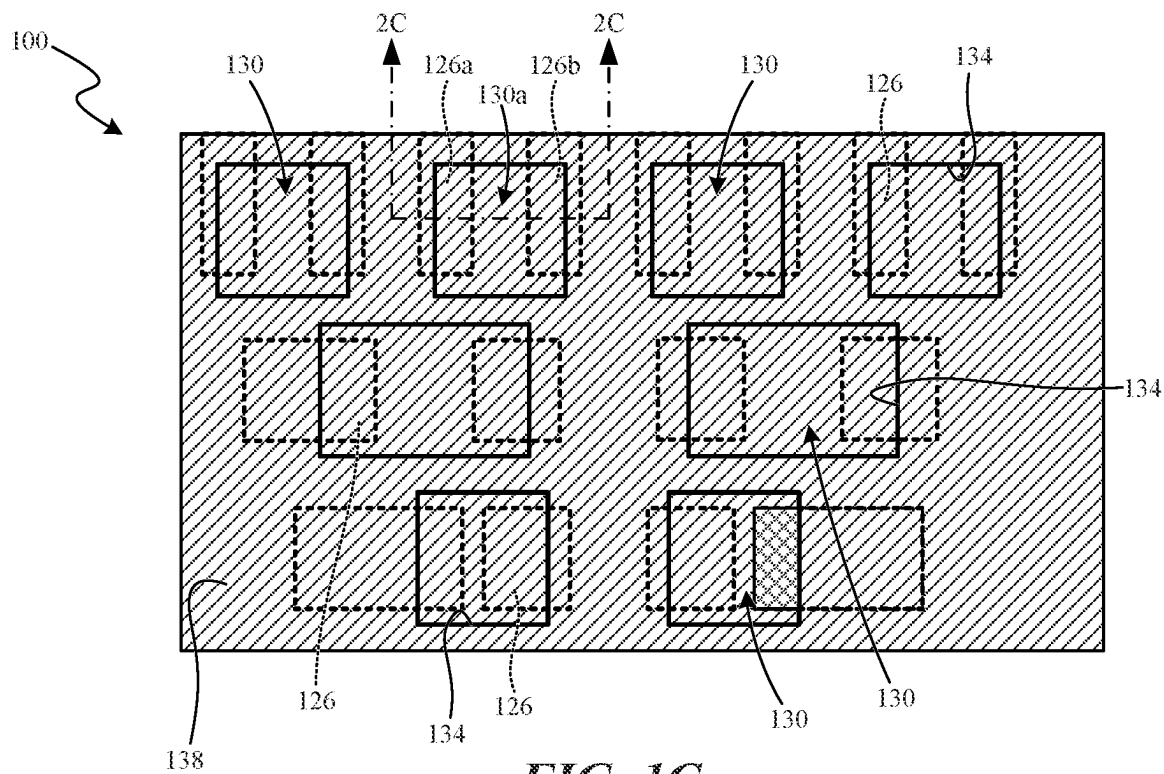


FIG. 1C

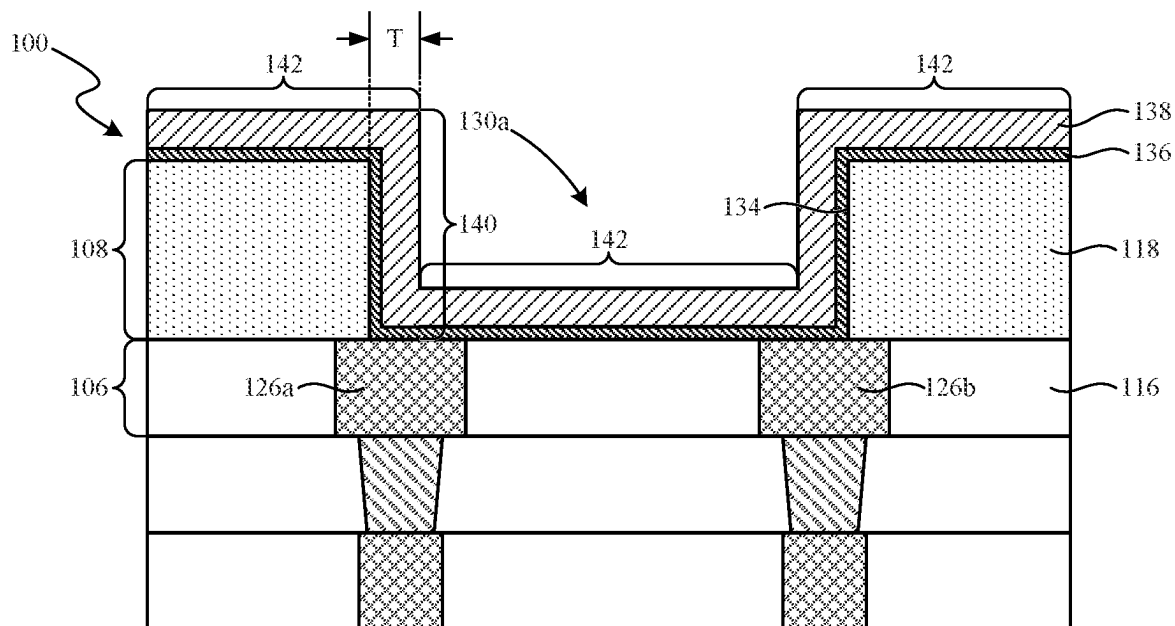


FIG. 2C

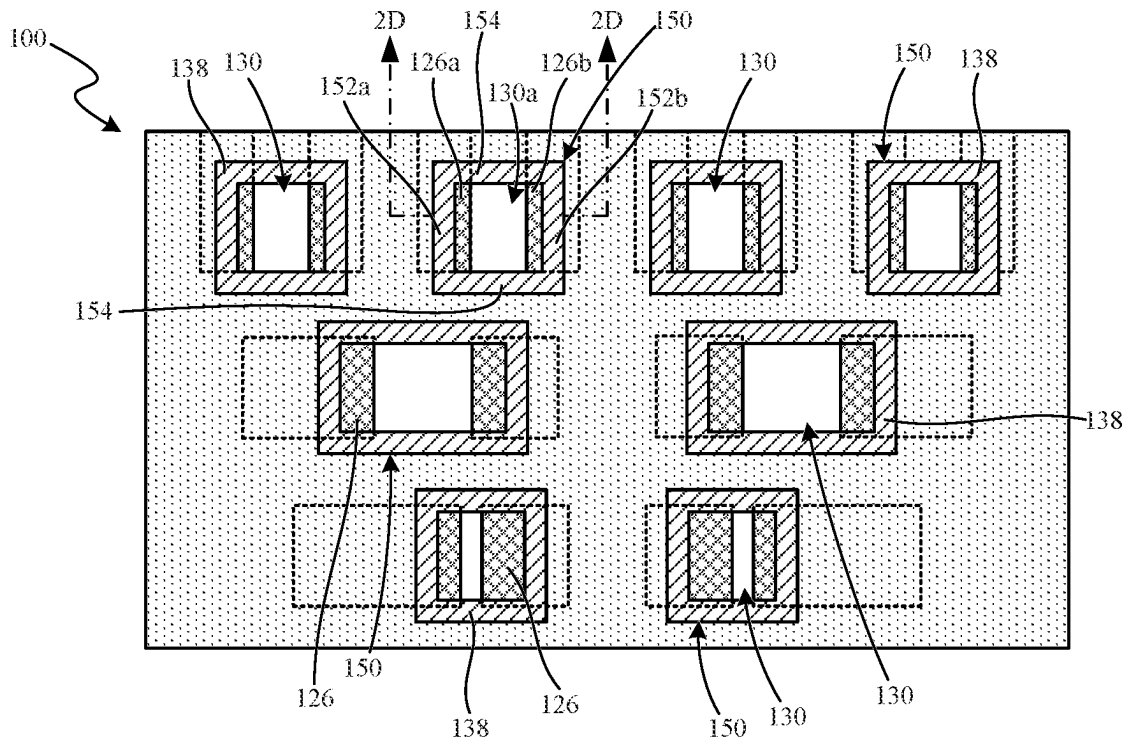


FIG. 1D

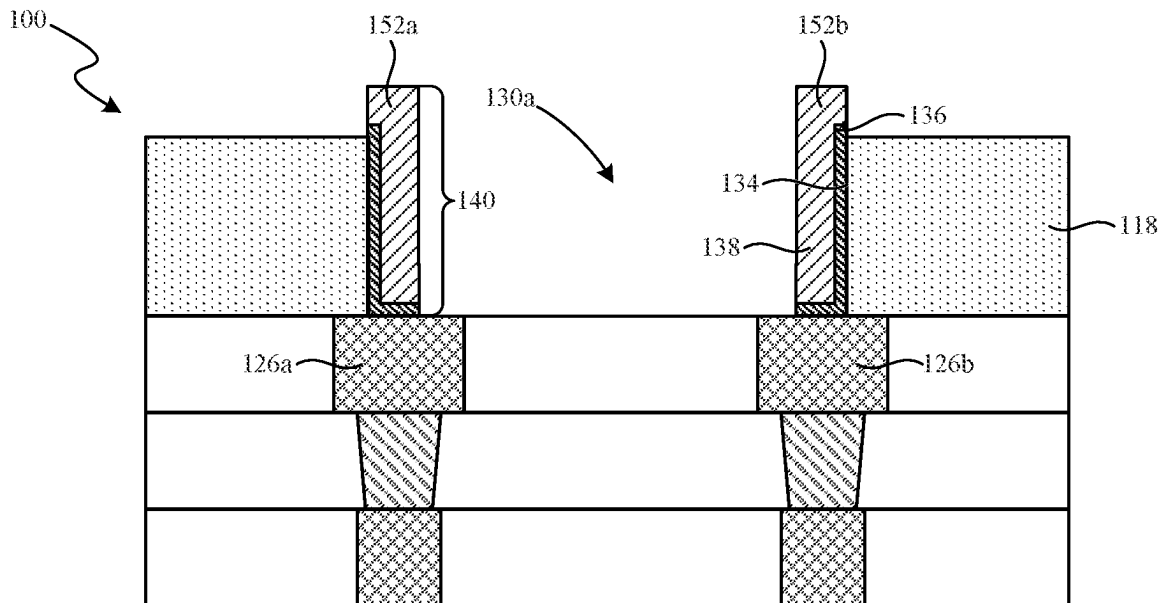


FIG. 2D

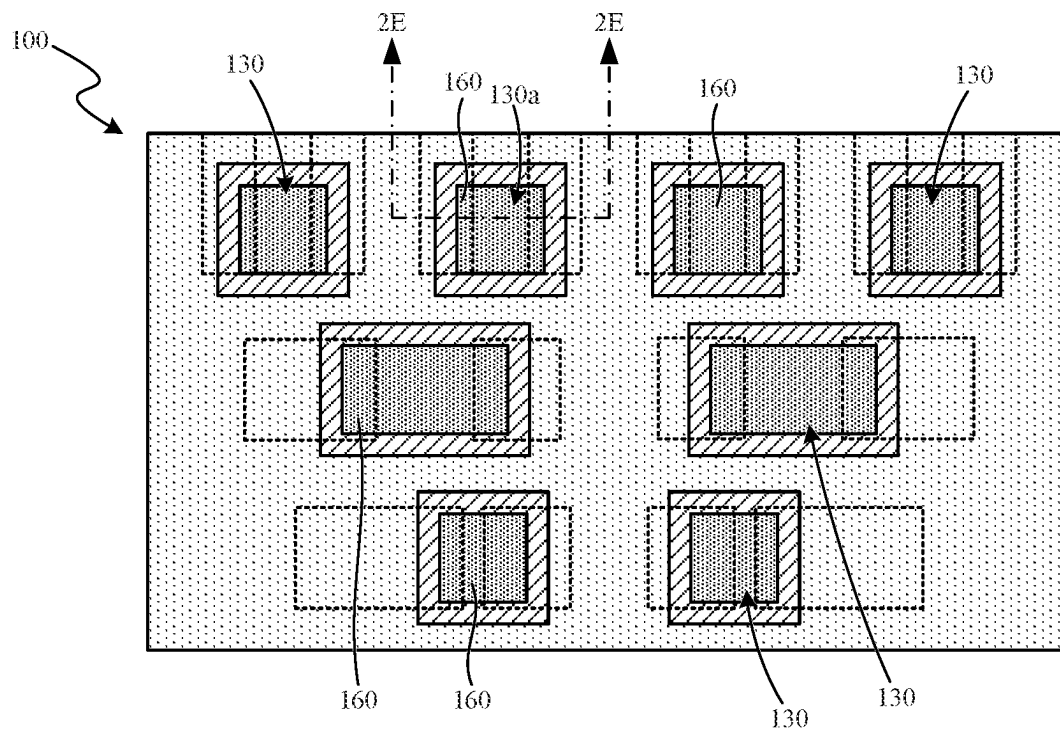


FIG. 1E

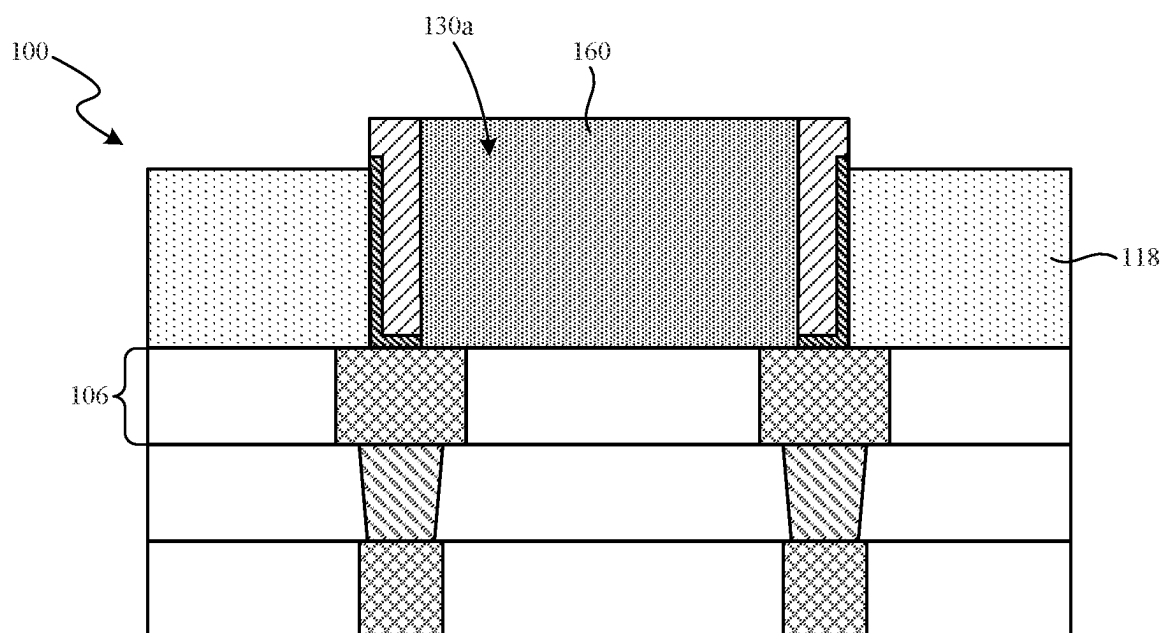


FIG. 2E

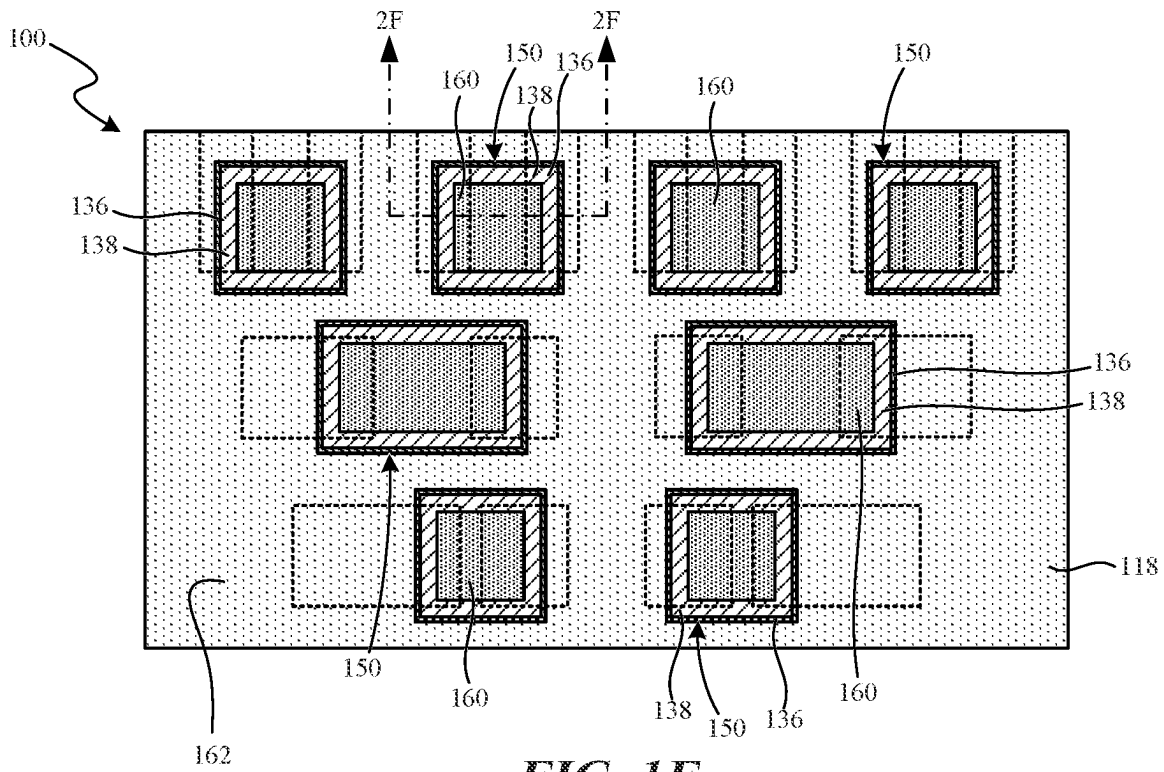


FIG. 1F

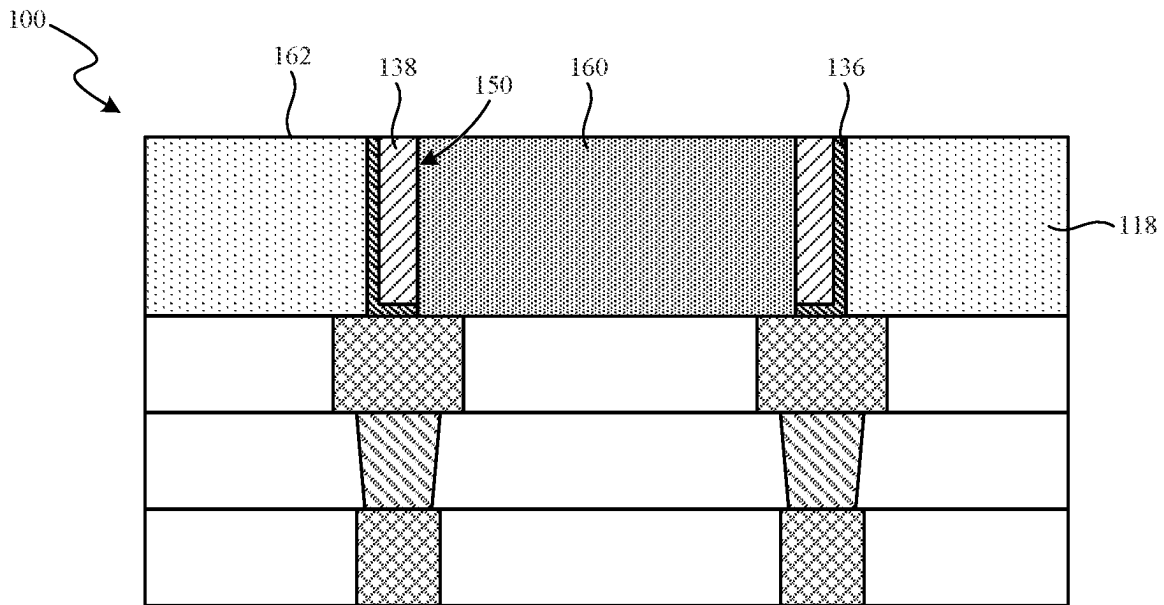


FIG. 2F

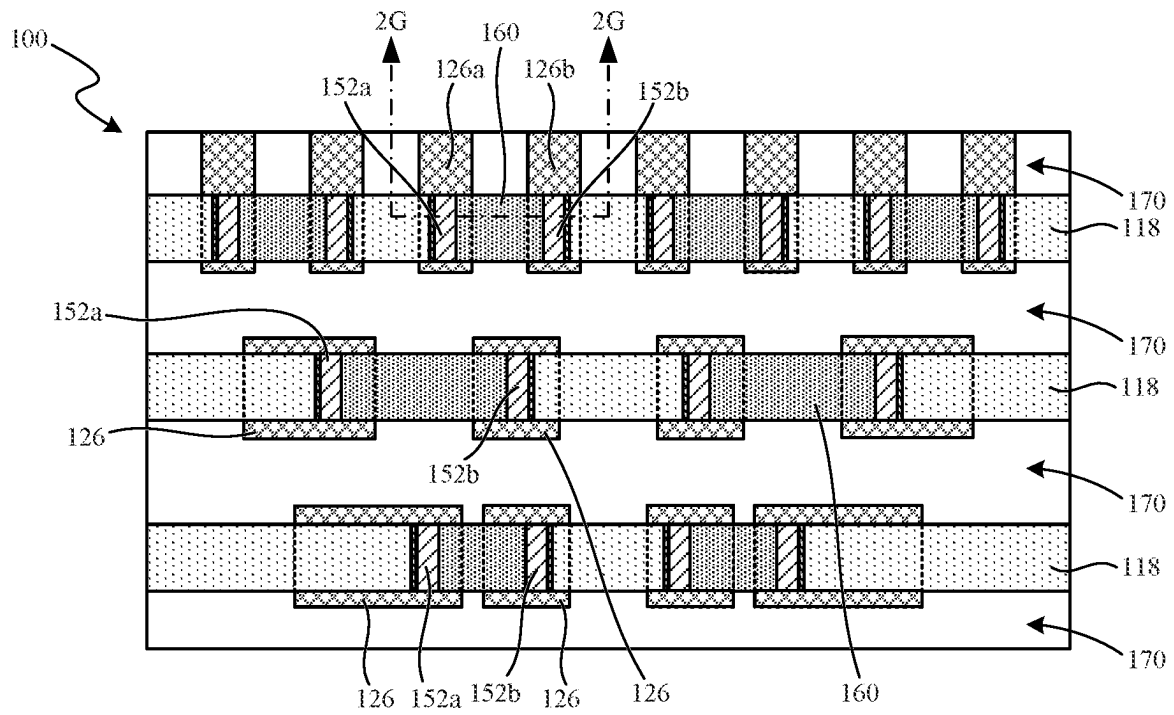


FIG. 1G

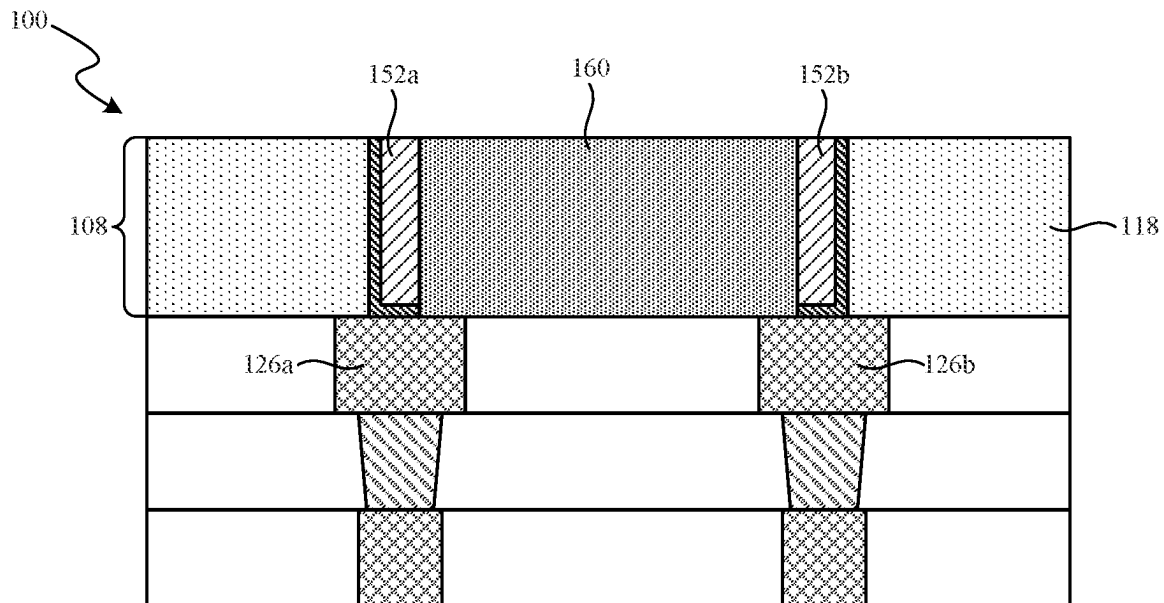


FIG. 2G

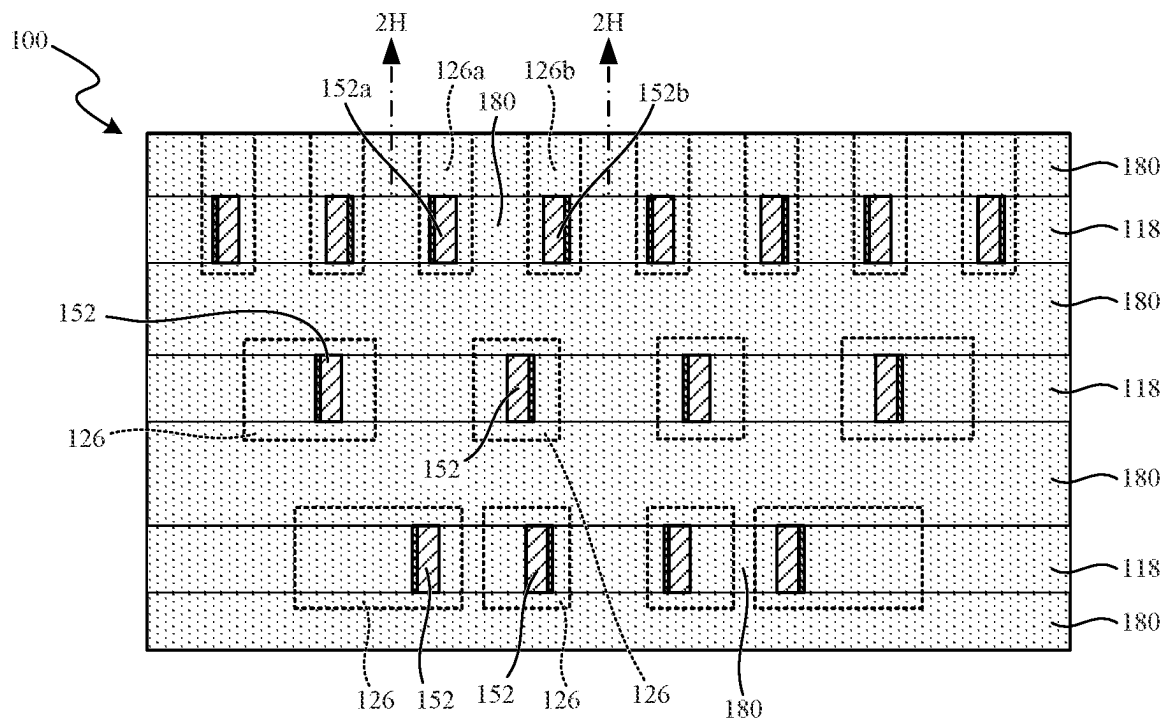


FIG. 1H

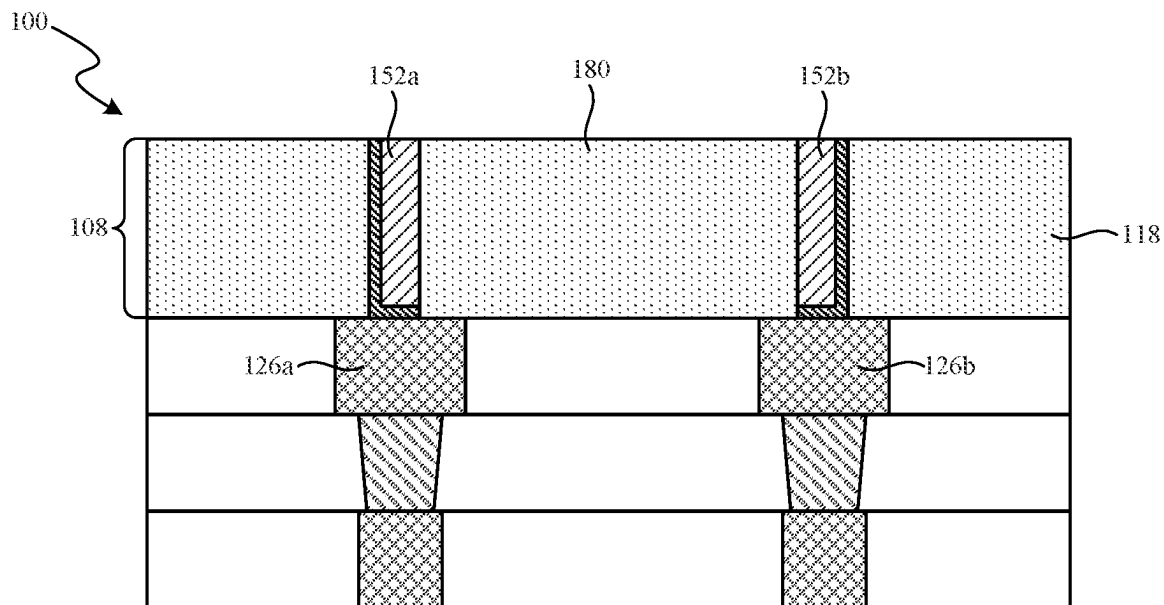


FIG. 2H

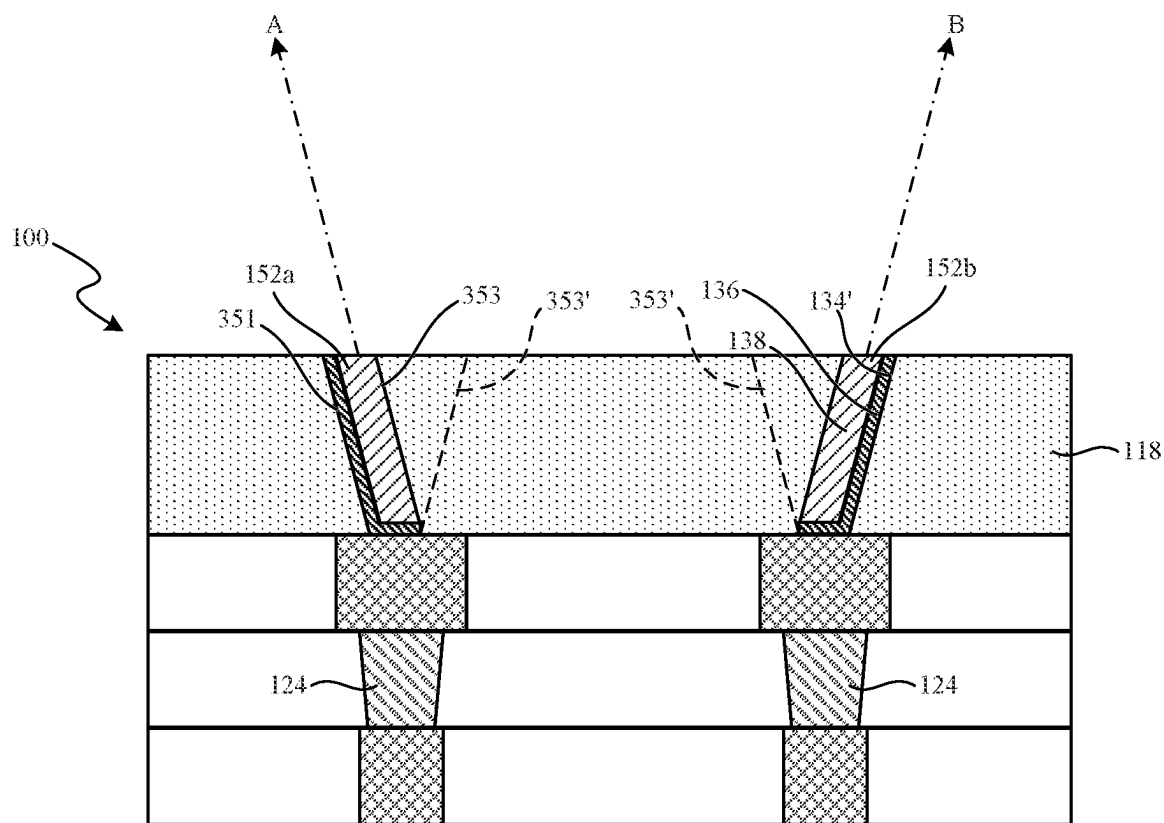
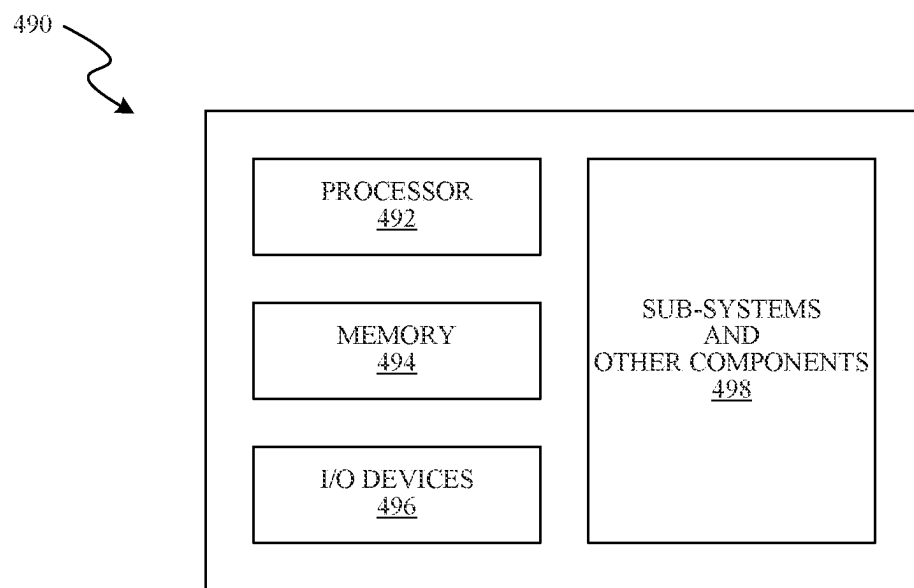


FIG. 3

***FIG. 4***

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METHODS FOR FORMING CONDUCTIVE VIAS, AND ASSOCIATED DEVICES AND SYSTEMS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 18/048,633, titled “METHODS FOR FORMING CONDUCTIVE VIAS, AND ASSOCIATED DEVICES AND SYSTEMS,” filed Oct. 21, 2022, which is a continuation of U.S. patent application Ser. No. 17/136,287, titled “METHODS FOR FORMING CONDUCTIVE VIAS, AND ASSOCIATED DEVICES AND SYSTEMS,” filed Dec. 29, 2020, now U.S. Pat. No. 11,515,204, each of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

The present technology generally relates to semiconductor devices and methods for manufacturing semiconductor devices, and more particularly relates to methods for forming conductive vias in a semiconductor device.

BACKGROUND

Memory devices are widely used to store information related to various electronic devices such as computers, wireless communication devices, cameras, digital displays, and the like. Information is stored by programming different states of a memory cell. Various types of memory devices exist, such as non-volatile memory devices (e.g., NAND Flash memory devices) and volatile memory devices (e.g., dynamic RAM (DRAM), synchronous dynamic RAM (SDRAM), and the like).

Improving memory devices, generally, may include increasing memory cell density, increasing read/write speeds or otherwise reducing operational latency, increasing reliability, increasing data retention, reducing power consumption, or reducing manufacturing costs, among other metrics. One way of reducing manufacturing costs is to improve manufacturing processes to increase the margin of successfully manufactured devices. Manufacturers can improve the manufacturing margin by implementing processes that, for example, increase the consistency or tolerance of manufacturing steps (e.g., removal or deposition of materials), improve the scale of manufacturing, and so on.

BRIEF DESCRIPTION OF THE DRAWINGS

Many aspects of the present technology can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale. Instead, emphasis is placed on illustrating clearly the principles of the present technology.

FIGS. 1A-1H are enlarged partially-schematic top views illustrating various stages in a method of manufacturing a semiconductor device in accordance with embodiments of the present technology.

FIGS. 2A-2H are enlarged side cross-sectional views of the semiconductor device of FIGS. 1A-1H taken along the lines 2A-2A through 2H-2H shown in FIGS. 1A-1H, respectively, in accordance with embodiments of the present technology.

FIG. 3 is an enlarged side cross-sectional view of the semiconductor device of FIGS. 1A-1H taken along the line

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2H-2H shown in FIG. 1H in accordance with additional embodiments of the present technology.

FIG. 4 is a schematic view of a system that includes a semiconductor device in accordance with embodiments of the present technology.

DETAILED DESCRIPTION

Embodiments of the present technology include methods of manufacturing semiconductor devices, such as memory devices, and associated devices and systems. In several of the embodiments described below, for example, a method of manufacturing a semiconductor device includes forming an opening in an electrically insulative material at least partially over a first electrically conductive feature and a second electrically conductive feature. The first and second electrically conductive features can be metal lines, such as word and/or bit lines of a memory device. The electrically insulative material can include a sidewall at least partially defining the opening. The method can further include forming a ring of electrically conductive material on/about the sidewall of the insulative material. The ring of conductive material can include (a) a first via portion over the first electrically conductive feature, (b) a second via portion over the second electrically conductive feature, and (c) connecting portions extending between the first and second via portions. In some embodiments, the ring can have a generally rectangular shape. The method can further include removing the connecting portions of the ring to electrically isolate the first via portion from the second via portion. For example, channels can be etched along the insulative material to remove the connecting portions of the ring while leaving the first and second via portions.

In some aspects of the present technology, the conductive via portions can be formed at a lower cost and/or with higher margin than conventional techniques for forming conductive vias. For example, the opening formed in the insulative material can be significantly larger (e.g., having a lower aspect ratio) than the subsequently formed via portions. Accordingly, the opening can be formed via an etching or other process that is less precise—and thus more reliable and lower cost—than conventional methods that etch high aspect ratio holes that correspond to the subsequent dimensions of the conductive vias formed therein. In additional aspects of the present technology, the thickness of the conductive via portions can be precisely controlled and made arbitrarily small via a selected deposition technique used to deposit the conductive material along the sidewall of the insulative material.

Numerous specific details are disclosed herein to provide a thorough and enabling description of embodiments of the present technology. A person skilled in the art, however, will understand that the technology may have additional embodiments and that the technology may be practiced without several of the details of the embodiments described below with reference to FIGS. 1A-4. For example, some details of memory devices well known in the art have been omitted so as not to obscure the present technology. In general, it should be understood that various other devices and systems in addition to those specific embodiments disclosed herein may be within the scope of the present technology.

As used herein, the terms “vertical,” “lateral,” “upper,” “lower,” “above,” and “below” can refer to relative directions or positions of features in the semiconductor devices in view of the orientation shown in the Figures. For example, “upper” or “uppermost” can refer to a feature positioned closer to the top of a page than another feature. These terms,

however, should be construed broadly to include semiconductor devices having other orientations, such as inverted or inclined orientations where top/bottom, over/under, above/below, up/down, and left/right can be interchanged depending on the orientation.

A person skilled in the relevant art will recognize that suitable stages of the methods described herein can be performed at the wafer level or at the die level. Therefore, depending upon the context in which it is used, the term “substrate” can refer to a wafer-level substrate or to a singulated, die-level substrate. Furthermore, unless the context indicates otherwise, structures disclosed herein can be formed using conventional semiconductor-manufacturing techniques. Materials can be deposited, for example, using chemical vapor deposition, physical vapor deposition, atomic layer deposition, spin coating, and/or other suitable techniques. Similarly, materials can be removed, for example, using plasma etching, wet etching, chemical-mechanical planarization, or other suitable techniques. A person skilled in the relevant art will also understand that the technology may have additional embodiments, and that the technology may be practiced without several of the details of the embodiments described below with reference to FIGS. 1A-4.

FIGS. 1A-1H are enlarged partially-schematic top views illustrating various stages in a method of manufacturing a semiconductor device **100** (e.g., a memory device) in accordance with embodiments of the present technology. FIGS. 2A-2H are enlarged side cross-sectional views of the semiconductor device **100** taken along the lines 2A-2A through 2H-2H shown in FIGS. 1A-1H, respectively, in accordance with embodiments of the present technology. Generally, the semiconductor device **100** can be manufactured, for example, as a discrete device or as part of a larger wafer or panel. In wafer-level or panel-level manufacturing, a larger structure is formed before being singulated to form a plurality of individual structures. For ease of explanation and understanding, FIGS. 1A-2H illustrate the fabrication of a portion of a single semiconductor device **100**. However, one skilled in the art will readily understand that the fabrication of the semiconductor device **100** can be scaled to the wafer and/or panel level—that is, to include many more components so as to be capable of being singulated into two or more semiconductor devices—while including similar features and using similar processes as described herein.

FIGS. 1A and 2A illustrate the semiconductor device **100** after formation of (i) a first layer **102** including a first insulative material **112** and first conductive features **122** (e.g., a first metallization layer), (ii) a second layer **104** over the first layer **102** and including a second insulative material **114** and conductive vias **124** electrically coupled/connected to corresponding ones of the first conductive features **122**, (iii) a third layer **106** over the second layer **104** and including a third insulative material **116** and second conductive features **126** (e.g., a second metallization layer including an individually identified first one of the second conductive features **126a** and a second one of the second conductive features **126b**) electrically coupled/connected to corresponding ones of the conductive vias **124**, and (iv) a fourth layer **108** over the third layer **106** and including a fourth insulative material **118**. The second conductive features **126** are fully or partially obscured in FIGS. 1A-1H and therefore shown at least partially schematically.

The first through fourth insulative materials **112-118** (collectively “insulative materials **112-118**”) can comprise one or more electrically-insulative materials such as a passivation material, dielectric material, oxide (e.g., silicon

oxide), and/or tetraethyl orthosilicate (TEOS), and the insulative materials **112-118** can be the same or different. In some embodiments, the fourth insulative material **118** comprises TEOS and has a thickness of between about 2000-5000 angstroms (e.g., about 3500 angstroms). The thicknesses of the first through fourth layers **102-108** (collectively “layers **102-108**”) can be the same or different. The first conductive features **122**, the conductive vias **124**, and the second conductive features **126** can comprise a metal such as tungsten, a metal alloy, and/or a conductive-metal containing material, and can have the same or varying dimensions (e.g., widths, diameters) and/or arrangements. In some embodiments, the first conductive features **122** comprise copper and the second conductive features **126** comprise tungsten.

The first conductive features **122** can be electrically coupled to one or more circuit elements (e.g., wires, traces, interconnects, transistors; not shown) that can be formed in/on a substrate (not shown). The circuit elements can include, for example, memory circuits (e.g., dynamic random memory (DRAM) or other type of memory circuits), controller circuits (e.g., DRAM controller circuits), logic circuits, and/or other circuits. In some embodiments, the first conductive features **122** are electrically coupled to one or more complementary metal-oxide-semiconductor (CMOS) circuits.

The second conductive features **126** can be metal lines, contacts, traces, or the like extending through/along the third layer **106**. In some embodiments, for example, the second conductive features **126** can be word lines and/or bit lines electrically coupled to one or more memory elements (not shown). As best seen in FIG. 1A, the second conductive features **126** can have different planform shapes and/or dimensions. In the illustrated embodiment, for example, the second conductive features **126** each have a rectangular shape. However, in the illustrated embodiment the first and second ones of the second conductive features **126a, b** have the same dimensions, while a third one of the second conductive features **126c**, a fourth one of the second conductive features **126d**, a fifth one of the second conductive features **126e**, and so on have different dimensions. In the illustrated embodiment, the second conductive features **126** are generally arranged in rows. In other embodiments, the second conductive features **126** can have different shapes (e.g., circular, rectilinear, polygonal, irregular, and so on), different sizes, and/or can be arranged differently (e.g., in more or fewer rows, irregularly, in a grid, spaced closer or farther apart from one another).

FIGS. 1B and 2B illustrate the semiconductor device **100** after forming openings **130** (e.g., including an individually identified first opening **130a**) through the fourth insulative material **118** of the fourth layer **108** to expose a portion of two or more of the second conductive features **126** in each of the openings **130**. For example, a portion of the first one of the second conductive features **126a** and a portion of the second one of the second conductive features **126b** are exposed in the first opening **130a**. A photolithography, etching, punching, chopping, masking, and/or other suitable process can be used to remove portions of the fourth insulative material **118** to form the openings **130**. In some embodiments, one or more additional layers **132** (shown schematically) can optionally be formed over the fourth layer **108** before forming the openings **130**. The additional layers **132** can comprise one or more resist, antireflective coating (ARC), backside antireflective coating (BARC), carbon, and/or other suitable layers known in the art of photolithography. In the illustrated embodiment, a portion of

adjacent ones of the second conductive features **126** are exposed in corresponding ones of the openings **130**. In other embodiments, some or all the openings **130** can be formed over (i) the entire footprint of one or more of the second conductive features **126**, (ii) more or less of the area of the second conductive features **126**, (iii) more than two of the second conductive features **126**, and/or (iv) non-adjacent ones of the second conductive features **126**.

Referring to FIG. 2B, in some embodiments the fourth insulative material **118** includes a generally vertical sidewall **134** at/defining each of the openings **130**. In other embodiments, the sidewalls **134** can be angled/slanted (e.g., relative to an upper surface of the third layer **106**) as shown in dashed lines in FIG. 2B and identified by reference numeral **134'**. In such embodiments, the openings **130** can have a tapered shape including, for example, a cross-sectional dimension (e.g., radius, diameter, area) that decreases in a direction toward the third layer **106**. Some such embodiments are described in further detail with reference to FIG. 3.

FIGS. 1C and 2C illustrate the semiconductor device **100** after forming/depositing (i) a seed layer **136** over the fourth insulative material **118** and in the openings **130** over the third layer **106** and (ii) a metal layer **138** over the seed layer **136**. The metal layer **138** can comprise a metal such as tungsten, copper, silver, aluminum, a metal alloy, a conductive-metal containing material, or the like, and is electrically coupled to the portions of the second conductive features **126** exposed in the openings **130** (e.g., via the seed layer **136**). In some embodiments, the seed layer **136** can comprise titanium nitride (TiN). The seed layer **136** and the metal layer **138** can be formed via sputtering, chemical vapor deposition, physical vapor deposition, atomic layer deposition, spin coating, electroplating, electro-less plating, and/or another suitable deposition technique. In some embodiments, the seed layer **136** and the metal layer **138** can be formed only in the openings **130** and not over the fourth insulative material **118**. In some embodiments, the seed layer **136** can be omitted or another suitable metallization structure can be formed.

Referring to FIG. 2C, the seed layer **136** and the metal layer **138** in each of the openings **130** can include a vertical portion **140** formed along the sidewall **134** of the opening **130** and a horizontal portion **142** formed over (i) the fourth insulative material **118** (e.g., an upper surface of the fourth insulative material **118**) and (ii) the third layer **106** exposed in the opening **130**. As described in detail below, the vertical portion **140** can be patterned to form individual conductive vias electrically coupled to the second conductive features **126** in the third layer **106**. Accordingly, a thickness **T** of the seed layer **136** and the metal layer **138** can be selected based on a desired final thickness of the conductive vias. In some embodiments, the thickness **T** can be less than 100 nanometers, less than 10 nanometers, less than 5 nanometers, less than 1 nanometer, or greater than 100 nanometers.

FIGS. 1D and 2D illustrate the semiconductor device **100** after removing the horizontal portions **142** (FIG. 2C) of the seed layer **136** and the metal layer **138** to leave only the vertical portions **140**. In some embodiments, a photolithography, punching, plasma etching, wet etching, and/or other suitable process can be used to remove the horizontal portions **142**. In some embodiments, the horizontal portions **142** can be removed using a straight punch process. As best seen in FIG. 1D, after removing the horizontal portions **142**, the seed layer **136** and the metal layer **138** can form a plurality of rings **150** extending along/about the sidewalls **134** of corresponding ones of the openings **130**.

With reference to the ring **150** formed in the first opening **130a**, the rings **150** can each include via portions **152** (e.g., first or vertical side or edge portions; identified individually as a first via portion **152a** and a second via portion **152b**) each positioned at least partially over a corresponding one of the second conductive features **126**. Specifically, the first via portion **152a** can be positioned partially over the first one of the second conductive features **126a** and the second via portion **152b** can be positioned partially over the second one of the second conductive features **126b**. Referring to FIG. 1D, the rings **150** can further include connecting portions **154** (e.g., second or horizontal side or edge portions) extending between and connecting the via portions **152**. In the illustrated embodiment, the rings **150** have a generally rectangular shape formed by the opposing via portions **152** and the opposing connecting portions **154**. In other embodiments, the rings **150** can have other shapes determined by, for example, the shape and dimensions of the openings **130**.

FIGS. 1E and 2E illustrate the semiconductor device **100** after depositing an under layer **160** in the openings **130** over the third layer **106**. The under layer **160** can comprise an oxide, a photoresist material, carbon-based spin on material, and/or another electrically-insulative material. In some embodiments, the under layer **160** is a sacrificial material configured to be removed during subsequent downstream processing steps. In other embodiments, the under layer **160** is configured to remain in the semiconductor device **100** after manufacturing and therefore can be, for example, the same material as the fourth insulative material **118**. The under layer **160** can be deposited via a spin-on process or another suitable deposition process.

FIGS. 1F and 2F illustrate the semiconductor device **100** after planarizing an upper surface **162** of the semiconductor device **100** including (i) the under layer **160** and (ii) the rings **150** formed from the seed layer **136** and the metal layer **138**. In some embodiments, the upper surface **162** can be planarized/polished to remove portions of the under layer **160** and the rings **150** using plasma etching, wet etching, chemical-mechanical planarization (CMP), buffing, and/or other suitable techniques.

FIGS. 1G and 2G illustrate the semiconductor device **100** after removing portions of the rings **150** (FIGS. 1F and 2F) and the fourth insulative material **118** in the fourth layer **108** to form channels **170**. More specifically, with additional reference to FIGS. 1D and 2D, the channels **170** can be formed along the semiconductor device **100** to remove the connecting portions **154** of the rings **150**—thereby electrically disconnecting the via portions **152** and leaving all or a portion of the via portions **152** on and electrically coupled to corresponding ones of the second conductive features **126**. In the illustrated embodiment, the channels **170** are formed to extend (i) generally parallel to one another, (ii) in the same direction as the (e.g., elongate) connecting portions **154**, and (iii) over multiple ones of the connecting portions **154** including, for example, the connecting portions **154** of multiple vertically and horizontally adjacent ones of the rings **150**. That is, the channels **170** are formed to remove the connecting edges of the rings **150** to electrically separate the via portions **152** of each of the rings **150**.

In other embodiments, the channels **170** can have different shapes, dimensions, and/or orientations based on, for example, the layout of the rings **150** and the second conductive features **126**. For example, some or all the channels **170** can be formed over/on (i) only a single one of the connecting portions **154** (e.g., rather than multiple adjacent or linearly aligned ones of the connecting portions) and/or (ii) non-adjacent and/or non-aligned ones of the connecting

portions **154**. The channels **170** can be formed using a photolithography, etching, punching, chopping, masking, and/or other suitable process to remove portions of the fourth layer **108**.

FIGS. **1H** and **2H** illustrate the semiconductor device **100** after removing the under layer **160** (FIGS. **1G** and **2H**) and depositing a fifth insulative material **180** into (i) the openings **130** (FIGS. **1E** and **2E**) previously filled with the under layer **160** and (ii) the channels **170** (FIGS. **1G** and **2G**). The fifth insulative material **180** can be deposited via a spin-on process or another suitable deposition process. In some embodiments, the fifth insulative material **180** is the same material as the fourth insulative material **118**, such as TEOS. Accordingly, after depositing the fifth insulative material **180**, each of the via portions **152** (i) extends through and is surrounded by the fourth and fifth insulative materials **118**, **180** and (ii) is electrically coupled to a corresponding one of the second conductive features **126**.

In some embodiments, the semiconductor device **100** can be planarized after deposition of the fifth insulative material **180**. In some embodiments, a metallization layer can subsequently be formed over the fourth layer **108**. The metallization layer can include, for example, metal lines (e.g., word and/or bit lines) or other conductive features electrically coupled to corresponding ones of the via portions **152**. Accordingly, the via portions **152** can comprise conductive vias electrically coupling the second conductive features **126** to a metallization layer over the fourth layer **108**.

In some aspects of the present technology, the conductive via portions **152** can be formed through the fourth layer **108** at a lower cost and/or with higher margin than conventional techniques for forming conductive vias. For example, with reference to FIGS. **1A-2H** together, the openings **130** are significantly larger than the subsequently formed via portions **152**. That is, the aspect ratio of the openings **130** is less than the aspect ratio of the subsequently formed via portions **152**. Accordingly, the openings **130** can be formed with an etching or other process that is less precise—and thus more reliable and lower cost—than conventional methods that etch high aspect ratio holes that correspond to the subsequent dimensions of the conductive vias formed therein. In additional aspects of the present technology, the thickness **T** of the conductive via portions **152** can be precisely controlled and made arbitrarily small via the deposition technique used to deposit the seed layer **136** and the metal layer **138**.

Referring again to FIG. **2B**, in some embodiments the etching process used to form the openings **130** can result in the sidewall **134'** of the fourth insulative material **118** being angled. FIG. **3** is an enlarged side cross-sectional view of the semiconductor device **100** taken along the line **2H-2H** shown in FIG. **1H** and illustrating the conductive via portions **152** formed along the angled sidewall **134'** in accordance with additional embodiments of the present technology. In the illustrated embodiment, the conductive via portions **152** extend generally parallel to the sidewall **134'** because, for example, the seed layer **136** and the metal layer **138** are directly deposited onto the sidewall **134'** (e.g., as shown in FIGS. **1C** and **2C**). Moreover, each of the via portions **152** can include a first sidewall **351** and a second sidewall **353** opposite the first sidewall **351** and extending generally parallel to the first sidewall **351**.

In contrast, conductive vias formed conventionally by etching high-aspect ratio holes typically include sidewalls that are slanted relative to one another (e.g., similar to the conductive vias **124** and/or as shown in dashed lines with reference to a sidewall **353'**)—as the conductive vias are

formed in the entire etched hole rather than on a single side of the hole. Because the conductive via portions **152** of the present technology are formed in the same first opening **130a** (FIGS. **1B** and **2B**; e.g., on opposite sides of the same first opening **130a**), the first and second sidewalls **351**, **353** can extend generally parallel to one another despite the angling of the sidewall **134'**. Moreover, because the via portions **152** are formed on opposite sides of the same one of the openings **130**, the conductive portions **152** in each of the openings **130** can slant in different directions. In the illustrated embodiment, for example, the first via portion **152a** is angled along a first axis **A** and the second via portion **152b** is angled along a second axis **B** extending in a different (e.g., opposite) direction as the first axis **A**.

In other embodiments, methods in accordance with the present technology can be used to form one, or more than one, conductive via in an opening formed in an insulative material over conductive contacts or lines. For example, with reference to FIGS. **1A-2H** together, the second one of the second conductive features **126b** could be omitted from the third layer **106**, but the fabrication process could proceed similarly to form the first opening **130a**, deposit the seed and metal layers **136**, **138**, and so on. Therefore, the second via portion **152b** can be formed over a portion of the insulative third layer **106** and, in some embodiments, can subsequently be removed (e.g., as part of the removal process shown in FIGS. **1G** and **2G** or a separate removal process). Accordingly, the first via portion **152a** can be formed to have the same high aspect ratio and with the same advantages as described above without requiring the simultaneous formation of a conductive via over an adjacent or nearby one of the second conductive features **126b**.

Similarly, the openings **130** can be formed over more than one of the second conductive contacts **126** (e.g., more than two). In some such embodiments, one or both of the connecting portions **154** of the rings **150** can be formed to conductively contact a corresponding one of the conductive contacts **126**. Then, the via portions **152** and the connecting portions **154** can be electrically disconnected using one or more etching/removal processes. In some embodiments, channels can be etched in different (e.g., orthogonal) directions to electrically isolate the via portions **152** and the connecting portions **154**.

The semiconductor device **100** described in detail above with reference to FIGS. **1A-3** and/or packages incorporating the semiconductor device **100** can be incorporated into any of a myriad of larger and/or more complex systems, a representative example of which is system **490** shown schematically in FIG. **4**. The system **490** can include a processor **492**, a memory **494** (e.g., SRAM, DRAM, flash, and/or other memory devices), input/output devices **496**, and/or other subsystems or components **498**. The memory devices and/or packages described above with reference to FIGS. **1A-3** can be included in any of the elements shown in FIG. **4**. The resulting system **490** can be configured to perform any of a wide variety of suitable computing, processing, storage, sensing, imaging, and/or other functions. Accordingly, representative examples of the system **490** include, without limitation, computers and/or other data processors, such as desktop computers, laptop computers, Internet appliances, hand-held devices (e.g., palm-top computers, wearable computers, cellular or mobile phones, personal digital assistants, music players, etc.), tablets, multi-processor systems, processor-based or programmable consumer electronics, network computers, and minicomputers. Additional representative examples of the system **490** include lights, cameras, vehicles, etc. With regard to these

and other example, the system 490 can be housed in a single unit or distributed over multiple interconnected units, for example, through a communication network. The components of the system 490 can accordingly include local and/or remote memory storage devices and any of a wide variety of suitable computer-readable media.

From the foregoing, it will be appreciated that specific embodiments of the technology have been described herein for purposes of illustration, but that various modifications may be made without deviating from the disclosure. Accordingly, the invention is not limited except as by the appended claims. Furthermore, certain aspects of the new technology described in the context of particular embodiments may also be combined or eliminated in other embodiments. Moreover, although advantages associated with certain embodiments of the new technology have been described in the context of those embodiments, other embodiments may also exhibit such advantages and not all embodiments need necessarily exhibit such advantages to fall within the scope of the technology. Accordingly, the disclosure and associated technology can encompass other embodiments not expressly shown or described herein.

We claim:

1. A method of manufacturing a semiconductor device, the method comprising:

forming a layer of electrically conductive material on at least a portion of a sidewall that at least partially defines an opening in an insulative material, wherein the opening is at least partially over a first electrically conductive feature and a second electrically conductive feature, and wherein the layer of electrically conductive material includes (a) a first via portion over the first electrically conductive feature, (b) a second via portion over the second electrically conductive feature, and (c) a connecting portion extending between the first via portion and the second via portion; and

removing the connecting portion of the layer of electrically conductive material to electrically isolate the first via portion from the second via portion.

2. The method of claim 1 wherein the method further comprises forming the opening in the insulative material.

3. The method of claim 1 wherein forming the layer of electrically conductive material includes forming the layer of electrically conductive material as a ring extending completely about the sidewall.

4. The method of claim 3 wherein the connecting portion comprises a first wall portion and a second wall portion opposite the first wall portion, and wherein removing the connecting portion comprises removing the first wall portion and the second wall portion.

5. The method of claim 1 wherein forming the layer of electrically conductive material comprises:

depositing the layer of the electrically conductive material in the opening over (a) the at least portion of the sidewall and (b) an insulative layer having the first and second electrically conductive features therein; and removing the electrically conductive material over the insulative layer.

6. The method of claim 5 wherein removing the electrically conductive material over the insulative layer includes straight punching the electrically conductive material over the insulative layer.

7. The method of claim 1 wherein the method further comprises depositing an under layer into the opening after forming the layer of electrically conductive material.

8. The method of claim 7 wherein the method further comprises planarizing the under layer and the layer of electrically conductive material.

9. The method of claim 7 wherein the insulative material is a first insulative material, and wherein the method further comprises:

removing the under layer after removing the connecting portion of the layer of electrically conductive material; and

depositing a second insulative material in place of the under layer.

10. The method of claim 9 wherein the first and second insulative materials are the same.

11. The method of claim 1 wherein forming the layer of electrically conductive material includes forming the layer of electrically conductive material to have a generally rectangular cross-sectional shape having opposing first sides and opposing second sides.

12. The method of claim 11 wherein the connecting portion extends along the first sides, and wherein the via portions extend along the second sides.

13. The method of claim 1 wherein forming the layer of electrically conductive material includes depositing the electrically conductive material onto the at least portion of the sidewall such that the electrically conductive material has a thickness of less than about 10 nanometers.

14. A method of manufacturing a semiconductor device, the method comprising:

depositing a layer of conductive material at least partially in a plurality of openings in an insulative material such that the conductive material forms rings in corresponding ones of the openings, wherein individual ones of the openings are positioned over a first electrically conductive feature and a second electrically conductive feature, and wherein individual ones of the rings include (a) a first via portion over the first conductive feature, (b) a second via portion over the second conductive feature, and (c) a connecting portion extending between the first via portion and the second via portion; and

removing the connecting portions of the rings of electrically conductive material to electrically isolate the first via portion from the second via portion of individual ones of the rings.

15. The method of claim 14 wherein the method further comprises forming the openings in the insulative material.

16. The method of claim 14 wherein the plurality of rings includes a first ring and a second ring, wherein the connecting portion of the first ring is adjacent the connecting portion of the second ring, and wherein removing the connecting portions includes etching a channel along an axis extending between the connecting portion of the first ring and the connecting portion of the second ring.

17. The method of claim 14 wherein removing the connecting portions of the rings of electrically conductive material includes etching a plurality of channels in the layer of insulative material.

18. The method of claim 17 wherein the channels extend generally parallel to one another.

19. A method of manufacturing a semiconductor device, the method comprising:

forming a ring of electrically conductive material on at least a portion of a sidewall that at least partially defines an opening in an insulative material, wherein the opening is at least partially over a first electrically conductive feature and a second electrically conductive feature, and wherein the ring of electrically conductive

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material includes (a) a first via portion over the first electrically conductive feature, (b) a second via portion over the second electrically conductive feature, and (c) a connecting portion extending between the first via portion and the second via portion; and
removing the connecting portion of the ring of electrically conductive material to electrically isolate the first via portion from the second via portion.

20. The method of claim **19** wherein forming the ring of electrically conductive material includes forming the ring of electrically conductive material to have a generally rectilinear cross-sectional shape having opposing first sides and opposing second sides, and wherein the connecting portion extends along the first sides, and wherein the first and second via portions extend along the second sides.

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